Saithan Panchaharan

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Education

University of Toronto Sep. 2024 – June 2029 *(Expected)*
BSc in Computer Science (Co-Op) *Toronto, ON*

- **Coursework**: *Software Engineering, Data Structures & Algorithms, Artificial Intelligence, Microprogramming, Assembly Programming, Operating Systems, Shell Programming, OOP, Statistics & Data Analysis*

Experience

Bellwoods Strategic Capital Sep. 2026 – Dec. 2026
Full-Stack Developer *Toronto, ON*

- Implementing features & fixing bugs in **Thrive: Happy Habits**, a **React Native**, **Expo** and **Django REST Framework** app (∼90 API routes, 16 Jest suites)
- Scoped **5** MVP user stories with the client in team meetings, prototyping user flows in **Figma**, & rebuilding the quest filtering logic
- Delivering biweekly **Android APK** builds for client review, targeting **Google Play** and **iOS App Store** releases

Projects

Sportfolio | *React, TypeScript, Google OAuth, Vercel* *Live Demo*

- Built a trading platform where users trade MLB/NFL players like stocks, using a **Spring Boot** pricing engine that blends 4 weighted signals into prices for **1,800+** players capped at **30%** daily movements
- Reduced daily repricing queries by **99%**, by batching **Spring Data JPA** queries & grouping results into per-player maps in memory using **Java Streams**
- Cut **Tank01** API calls to zero for historical repricing by archiving raw game stats in **PostgreSQL**, & cached player names, projections, & draft rankings for 24 hours
- Replaced a 1,900-parameter IN clause with a single date-bounded query, eliminating production database connection drops on **Railway** & keeping query size constant as the roster grows

LineTracker | *SQLAlchemy, yfinance, Google OAuth, Brevo* *Live Demo*

- Built a full-stack platform with a **React** web app and **Flask** API that emails users the moment a stock price or sports betting line hits their target
- Reduced upstream API calls by **90%** by grouping active alerts by sport, market, & ticker, scaling **SharpAPI** & stock-price requests with distinct games & tickers, rather than alert count
- Checked every active alert every **90** seconds without breaching SharpAPI's free-tier limit of about 12 requests/minute, using **APScheduler**, **6**-second request spacing, & a **150**-second odds cache
- Shipped a native iOS app in **SwiftUI** on the same **Flask** and **PostgreSQL** backend, with **Keychain**-backed sessions & alerts to sync with the web app

YouTube Video Summarizer | *Python, JavaScript, Vite, Gunicorn* *Live Demo*

- Built a web app that summarizes YouTube videos at 3 expertise levels, using **Claude API** on transcripts fetched
- Cut requests when switching expertise levels by **50%** by caching the transcript client-side & only re-running the summary endpoint
- Restored transcript fetching in production by switching to the **Supadata API** after YouTube blocked requests from cloud IPs
- Deployed the **React** frontend on **Vercel** & **Flask** backend on **Render** with environment-based API keys & CORS configuration, adding read-aloud playback using **Web Speech API**

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C/C++, C#, Go, Kotlin, Swift, SQL, Bash, HTML5, CSS3
Frameworks: React, Next.js, Angular, React Native, SwiftUI, Spring Boot, Node.js, Flask, FastAPI, Django REST
Tools: PostgreSQL, MongoDB, Redis, Kafka, AWS, Azure, Docker, Kubernetes, Terraform, Git, GitHub Actions